

DNA storage could make data centres obsolete

Imagine being able to store a data centre's worth of information on something the size of a single USB stick. A new technique developed by a team of computer scientists could one day make this possible using something borrowed from nature: DNA.

Researchers from the University of Washington and Microsoft Research have developed one of the first complete systems to encode, store and retrieve digital data using DNA molecules, capable of storing information millions of times more compactly than current data storage technologies.



DNA data storage (Image courtesy: <http://marinataracova.com>)

A paper presented recently at the Architectural Support for Programming Languages and Operating Systems (ASPLOS) conference in Atlanta describes how four images were successfully encoded into DNA snippets. More significantly, the researchers were able to reverse the process and perfectly retrieve the images without losing a single byte of information.

The images, which are encoded as a string of 0's and 1's, are converted into a string of A's, C's, T's and G's—the bases that pair to form DNA. A DNA molecule with that sequence is then chemically synthesised and dried out for storing with billions of other molecules.

"This works for any digital data, not just images," says Luis Ceze, an associate professor of electrical engineering at the University of Washington and one of the authors of the study. "We used images because images and video tend to take lots of storage space."

All the movies, images, emails and other digital data from more than 600 basic smartphones (10,000 gigabytes) can be stored in the faint pink smear of DNA at the end of a test tube. Digital data—including videos, photos and text—collected by devices is expected to hit 44 trillion gigabytes by 2020. At its current rate, the world is producing more data than the capacity to store it. Ceze and his colleagues believe DNA could be the solution to this problem.

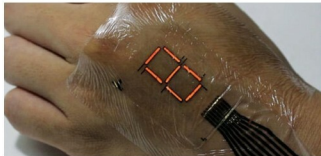
Ceze adds, "DNA also never becomes obsolete, unlike that old dusty floppy disk at the bottom of your drawer."

Electronic 'tattoo' transforms skin into a screen

Researchers have taken the concept of wearable technology to a whole new level by developing an electronic patch that transforms skin into an LED screen. The flexible display, created by scientists at the University of Tokyo, is as thin as cling film and is described as electronic skin, or e-skin.

The research, published in the journal *Science Advances*, describes how the technology could be used to display the wearer's blood oxygen level or heart rate, meaning potential applications could be both medical and consumer. To achieve the flexibility required for such a device, the team of scientists used alternating layers of inorganic (silicon oxynitride) and organic (polyene) material. This prevented oxygen and water vapour in the air from damaging the electronics.

Electronic displays have been stuck to skin before; however, the lifespan is usually only a few hours. In tests, the e-skin was able to last for several days.



Electronic skin tattoo (Image courtesy: www.newsweek.com)